Manuel Pérez Carrasco

Munich, Bavaria, Germany | (+56) 997946050
m.perez-carrasco@tum.de | Portfolio | Google Scholar

Research Interests

My research applies machine learning to Earth observations for environmental sensing and atmospheric science applications. I specialize in computer vision, time series analysis, and natural language processing.

Education

PhD Student, Environmental Sensing and Modeling — Munich, Germany
Technical University of Munich (TUM), Department of Electrical and Computer Engineering — *Oct. 2026 – Present*

- Supervisor: Prof. Dr.-Ing. Jia Chen, Professorship for Environmental Sensing and Modeling
- Thesis Project: *Integrated Artificial Intelligence System for the Detection and Quantification of Greenhouse and Trace Gas Emissions from Satellite Observations*

MSc. Computer Science — Concepción, Chile
University of Concepción (GPA: 6.1/7.0) — *Mar. 2017 – Oct. 2019*

- Thesis *Adversarial Variational Domain Adaptation for Semi-supervised Image Classification* (Grade: 7.0/7.0) received Best Student Paper Award at Astroinformatics 2019 Conference, Caltech
- Research conducted at Harvard University's Institute for Applied Computational Science with Professors Guillermo Cabrera-Vives and Pavlos Protopapas

Industrial Civil Engineering — Concepción, Chile
University of Concepción (GPA: 5.9/7.0) — *Mar. 2013 – Sept. 2018*

- Undergraduate honors thesis *Attribute transfer in deep neural networks applied to astronomical images* (Grade: 7.0/7.0) published in *Publications of the Astronomical Society of the Pacific*
- Focus on data science, machine learning, and optimization

Experience

SPARK Postbaccalaureate Research Fellow — Cambridge, MA, USA
Center for Astrophysics (CfA) | Harvard & Smithsonian — *Sept. 2025 – Sept. 2026*
Working under Dr. Cecilia Garraffo (Director, AstroAI at CfA) and Dr. Xiong Liu (PI of TEMPO, SAO Lead of MethaneSAT).

- **MethaneSAT Mission – Atmospheric Remote Sensing**: Developed deep learning algorithms for methane plumes, clouds and cloud shadows segmentation in spectroscopic imagery from MethaneSAT and MethaneAIR airborne data.
- **TEMPO Mission – Atmospheric Remote Sensing**: Designing geometry-based labeling system and machine learning models for cloud and shadow detection using spectroscopic data from NASA's Tropospheric Emissions: Monitoring of Pollution (TEMPO) satellite.

Chief Technology Officer — Concepción, Chile
Center for Data and Artificial Intelligence, University of Concepción — *Jul. 2019 – Aug. 2025*

- **Deep-Hub Geospatial Platform for Precision Silviculture**: Co-architected cloud-native ML infrastructure for automated tree detection using multi-sensor aerial imagery. Developed and deployed scalable deep learning models (detection, segmentation, land cover classification) processing 300,000+ acres of forest plantations. Led technical team of 5 engineers. Platform spun off as commercial venture from School of Engineering. Published research in *International Journal of Digital Earth*.
- **ALeRCE Anomaly Detector – Astronomical Time Series Analysis**: Designed and implemented deep learning-based anomaly detection algorithms identifying rare transient events in real-time telescope data. Research published in *The Astronomical Journal*. Funded by Millennium Institute of Astrophysics.
- **NLP for Social Science – Large-Scale Text Analysis**: Led development of transformer-based NLP platform for analyzing 500,000+ citizen feedback responses during Chilean social outbreak. Implemented sentiment analysis and topic modeling techniques, providing data-driven insights for policymakers. Platform's statistical outputs informed Chilean constitutional drafting process. Worked with Chilean Ministry of Social Development and Family.

Research Assistant Cambridge, MA, USA
Institute for Applied Computational Science, Harvard University *Dec. 2018 – May 2021*

- Conducted research on semi-supervised domain adaptation for image classification under Prof. Pavlos Protopapas (IACS Scientific Program Director). Developed novel contrastive learning framework (Con^{2}DA) for domain adaptation, published at NeurIPS 2021 Workshop on Distribution Shifts.
- Master's thesis research on adversarial variational domain adaptation. Recipient of IACS financial support (stipend and travel funding) for research development.
- Applied deep learning methods to astronomical image classification problems, transferring network knowledge across different telescope surveys and observing conditions.

Selected Publications

Pérez-Carrasco, M., et al. (2026). Plume Segmentation from MethaneSAT with Cross-Sensor Transfer Learning and Physics-Informed Postprocessing. *Preprint*. Link

Zhang, Z., et al., incl. **Pérez-Carrasco, M.**, Chen, J. (2026). Automatic methane plume masking based on wavelet transform image processing: application to MethaneAIR and MethaneSAT data. Atmospheric Measurement Techniques, 19, 4637–4651. Link

Pérez-Carrasco, M., et al. (2026). Deep learning for clouds and cloud shadow segmentation in methane satellite and airborne imaging spectroscopy. *IEEE Transactions on Geoscience and Remote Sensing*, 64, 1-20. Link

Pérez-Carrasco, M., et al. (2023). Alert classification for the ALeRCE broker system: The anomaly detector. *The Astronomical Journal*, 166(4), 151. Link

Pérez-Carrasco, M., et al. (2022). Precision silviculture: Use of UAVs and comparison of deep learning models for the identification and segmentation of tree crowns in pine crops. *International Journal of Digital Earth*, 15(1), 2223. Link

Sanchez-Saez, P., et al. (2021). Searching for changing-state AGNs in massive data sets. I. Applying deep learning and anomaly-detection techniques. *The Astronomical Journal*, 162(5), 206. Link

Förster, F., et al. (2021). The automatic learning for the rapid classification of events (ALeRCE) alert broker. *The Astronomical Journal*, 161(5), 242. Link

Pérez-Carrasco, M., et al. (2019). Multiband galaxy morphologies for CLASH: A convolutional neural network transferred from CANDELS. *Publications of the Astronomical Society of the Pacific*, 131(1004), 108002. Link

Conference Presentations

ICLR 2026: 4th ICLR Workshop on Machine Learning for Remote Sensing - Accepted Poster. Link
AAAI 2026: Workshop on AI to Accelerate Science and Engineering - Accepted Poster. Link
AGU 2025: AI-Driven Innovations in Earth and Atmospheric Sciences - Poster Presentation. Link
NeurIPS 2025: Climate Change ML Workshop – Poster Presenter. Link
NeurIPS 2023: Climate Change ML Workshop – Poster Presenter. Link
ICML 2023: Machine Learning for Astrophysics Workshop – (3) Poster Presentations. Link 1, Link 2, Link 3
NeurIPS 2021: Workshop on Distribution Shifts – Poster Presenter. Link
Astroinformatics 2019: Oral. Best Student Paper Award, Caltech. Link

Teaching Experience

Lecturer Concepción, Chile
School of Engineering, University of Concepción *Jul. 2019 – Dec. 2024*

- Taught graduate and undergraduate courses including Deep Learning (2022-2024), Introduction to Data Science (2021-2022), Machine Learning (2019-2020), and Data Analysis (2024). Co-designed curriculum, lectures, and assessments for classes of ∼30 students. Worked with professor Guillermo Cabrera-Vives.
- Topics covered: supervised/unsupervised learning, cross-validation, convolutional neural networks, recurrent architectures, transformers, computer vision, NLP and time series analysis.

Teaching Fellow Cambridge, MA, USA
Institute for Applied Computational Science, Harvard University *Feb. 2019 – May 2019*

- Teaching fellow for CS109b: Advanced Topics in Data Science, course for the Data Science Master's program. Conducted office hours, graded assessments, and mentored students on ML projects under Prof. Pavlos Protopapas.

Honors & Awards

- Visiting Professor, Master in Data Science for Innovation at University of Concepción (2023-2025)
- Best Student Paper Award, Astroinformatics 2019 Conference, Caltech
- UdeC Scholarship for Master studies (2017-2019)

Mentorship and Service

AstroAI/EarthAI: Affiliate, Center for Astrophysics | Harvard & Smithsonian (2023-2026).
NeurIPS 2024: LatinX in AI Workshop – Finance/Sponsorship Co-Chair
Master in Data Science for Innovation at UdeC: Mentored final graduation projects for 7 students.